Understanding Syntax: Python and R Compared: A Comparative Introduction for Data Analysis

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Abstract

This handout complements the presentation *Understanding Syntax: Python and R Compared*. It explains the most important syntax differences between R and Python for introductory data analysis. The focus is on variables, data structures, indexing, filtering, data manipulation, and translation between R's `dplyr` style and Python's `pandas` style. The aim is not to decide which language is better, but to help students understand how the same analytical logic can be expressed differently in each language.

Understanding Syntax: Python and R Compared: A Comparative Introduction for Data Analysis

1 Introduction

R and Python are two of the most widely used programming languages in data analysis. R was developed with a strong focus on statistical computing, data analysis, and visualisation. Python, by contrast, is a general-purpose programming language that has become highly important in data science through libraries such as `pandas`, `NumPy`, and `scikit-learn` (McKinney, 2022; Python Software Foundation, 2024).

For students in data analysis, the most important point is that both languages often solve the same analytical problem. The difference is mainly the syntax: how the instruction is written. Therefore, this handout explains the similarities and differences between the two languages using short practical examples.

The central learning objective is:

> Students should be able to read simple R and Python code, identify equivalent syntax patterns, and translate basic data manipulation operations between the two languages.

1.1 2. Python and R at a Glance

The following table summarises the most important differences.

Feature	R	Python
Main purpose	Statistical computing and graphics	General-purpose programming and data science
Main data structure	`data.frame` / `tibble`	`pandas.DataFrame`
Assignment	`x <- 10`	`x = 10`
First index	`1`	`0`
Popular data package	`dplyr` / `tidyverse`	`pandas`
Typical workflow	Pipe-based analysis	Method chaining / object-based analysis

Although the syntax looks different, the analytical thinking is often the same. For example, both languages can import data, clean data, filter rows, calculate summary statistics, and create visualisations. The user only needs to understand how each language expresses these tasks.

1.2 3. Assignment and Variables

A variable stores information that can be reused later. In R, the preferred assignment operator is `<-`, although `=` also works. In Python, assignment is done with `=`.

R

```
[1] "Srikumar"
```

Python

```
Srikumar
```

A small but important difference is that R uses uppercase logical values, such as `TRUE` and `FALSE`. Python uses `True` and `False`. Python is also stricter about indentation, especially in control structures such as `if` statements and loops.

1.3 4. Data Structures

Data structures organise information. In introductory data analysis, the most important structure is the data frame. A data frame is a table with rows and columns, similar to an Excel sheet.

R data frame

```
   name score   program
1 Alice    88   Finance
2   Bob    76 Marketing
3 Carol    95   Finance
```

Python DataFrame

```
    name  score    program
0  Alice     88    Finance
1    Bob     76  Marketing
2  Carol     95    Finance
```

In R, a data frame is available in base R. In Python, the standard data frame structure usually comes from the `pandas` library (The pandas Development Team, 2024). This means that Python users normally import pandas before creating or analysing tabular data.

1.4 5. Indexing: One-Based vs. Zero-Based Counting

One of the most common mistakes when moving between R and Python is indexing. R starts counting at 1, while Python starts counting at 0.

R

```
[1] 85
```

```
[1] 90
```

Python

```
85
```

```
90
```

This distinction is small, but it can create errors in practical analysis. When translating code, the index position must always be checked carefully.

1.5 6. Filtering and Selecting Data

Filtering means keeping only rows that satisfy a condition. Selecting means keeping only specific columns.

1.5.1 6.1 Filtering Rows

R with dplyr

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
   name score program
1 Alice    88 Finance
2 Carol    95 Finance
```

Python with pandas

```
    name  score  program
0  Alice     88  Finance
2  Carol     95  Finance
```

In R, `filter()` is commonly used in the `dplyr` package. In Python, row filtering often uses Boolean indexing, where the condition is placed inside square brackets.(Wickham & Grolemund, 2017)

1.5.2 6.2 Selecting Columns

R

```
   name score
1 Alice    88
2   Bob    76
3 Carol    95
```

Python

```
    name  score
0  Alice     88
1    Bob     76
2  Carol     95
```

The double square brackets in Python are important because a list of column names is passed into the DataFrame.

1.6 7. Data Manipulation with `dplyr` and `pandas`

R's `dplyr` package is popular because it makes code readable. It uses verbs such as `filter()`, `select()`, `mutate()`, `summarise()`, and `arrange()` (Wickham & Grolemund, 2017). Python's `pandas` library performs similar tasks, but the syntax is object-oriented.

Task	R / dplyr	Python / pandas
Add a column	`mutate()`	`.assign()` or `df["new"] = ...`
Filter rows	`filter()`	Boolean indexing or `.query()`

Task	R / dplyr	Python / pandas
Sort rows	`arrange()`	`.sort_values()`
Group data	`group_by()`	`.groupby()`
Calculate summary	`summarise()`	`.agg()` or `.mean()`

1.7 8. Complete Translation Example

The following example shows how the same task can be written in R and Python. The task is to calculate price per unit, keep only products with revenue above 10,000, and sort by revenue.

R Code

```
  product revenue units price_per_unit
1       C   22000   200       110.0000
2       A   15000   120       125.0000
3       D   11000    95       115.7895
```

Python Code

```
  product  revenue  units  price_per_unit
0       C    22000    200      110.000000
1       A    15000    120      125.000000
2       D    11000     95      115.789474
```

The example demonstrates that the analytical logic is identical. First, a new column is created. Second, rows are filtered. Third, the output is sorted. The main difference is how each language expresses these steps.

2 Conclusion

The comparison between R and Python shows that syntax is a surface-level difference, while the underlying data analysis logic is often the same. A student who understands how to filter, select, group, and summarise data in one language can transfer that knowledge to the other language.

The most important points are:

- R is especially strong for statistics, academic data analysis, and visualisation.
- Python is highly flexible and widely used for data science, automation, and machine learning.
- R starts indexing at 1; Python starts indexing at 0.
- R's pipe-based syntax is similar in purpose to pandas method chaining.
- Learning both languages improves analytical flexibility and makes collaboration easier.

For this course, R remains the primary language because the module focuses on data analysis and decision-making using R. However, understanding Python is useful because many business analytics environments use both tools.

3 Affidavit

I hereby affirm that this submitted paper was authored unaided and solely by me. Additionally, no other sources than those in the reference list were used. Parts of this paper, including tables and figures, that have been taken either verbatim or analogously from other works have in each case been properly cited with regard to their origin and authorship. This paper either in parts or in its entirety, be it in the same or similar form, has not been submitted to any other examination board and has not been published.

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3.1 Checklist

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- ☒ Either the handout or the presentation contains R code that demonstrates coding expertise.
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GitHub Repository: https://github.com/Sri-prog-7943/Python-and-R-Compared
[Srikumar Ravichandran,] [28 April 2026,] [Rösrath]

4 References

McKinney, W. (2022). *Python for data analysis: Data wrangling with pandas, NumPy, and jupyter* (3rd ed.). O'Reilly Media.

Python Software Foundation. (2024). *The python language reference*. https://docs.python.org/3/reference/.

The pandas Development Team. (2024). *Pandas documentation*. https://pandas.pydata.org/docs/.

Wickham, H., & Grolemund, G. (2017). *R for data science: Import, tidy, transform, visualize, and model data*. O'Reilly Media. https://r4ds.had.co.nz/